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MILITARY SPECIFICATION

# PUMPS, CENTRIFUGAL: 4-INCH AND 6-INCH, GASOLINE-ENGINE- AND DIESEL-ENGINE-DRIVEN, FOR WATER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers 4-inch and 6-inch self-priming centrifugal pumps for water pumping service.

1.2 Classification. The pumps shall be the following types, sizes, and classes, as specified (see 6.2):

Type I - Integral self-priming.
  Size 4 - 4-inch suction and discharge.
    Class A - Capacity as shown on figure 1.
    Class B - Capacity as shown on figure 2.
  Size 6 - 6-inch suction and discharge.
Type II - Auxiliary self-priming.
  Size 4 - 4-inch suction and discharge.

Beneficial comments (recommendations, additons, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Command, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| QQ-S-635 | - Steel Plate, Carbon. |
| VV-F-800 | - Fuel Oil, Diesel. |
| VV-G-1690 | - Gasoline, Automotive, Leaded or Unleaded. |
| WW-F-406 | - Flanges, Cast-Iron (Classes 125 and 250) and Bronze (Classes 150 and 300). |
| WW-P-471 | - Pipe-Fittings: Bushings, Locknuts, and Plugs; Iron, Steel, and Aluminum; (threaded); 125-150 Pounds. |

MILITARY

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| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| MIL-T-704 | - Treatment and Painting of Materiel. |
| MIL-L-2104 | - Lubricating Oil, Internal Combustion Engine, Tactical Service. |
| MIL-L-2105 | - Lubricating Oil, Gear, Multipurpose. |
| MIL-G-3056 | - Gasoline, Automotive, Combat. |
| MIL-F-3541 | - Fittings, Lubrication. |
| MIL-G-3859 | - Grease Guns, Hand-Operated, Lever, Push and Screw Type. |
| MIL-E-10062 | - Engines: Preparation for Shipment and Storage of. |
| MIL-P-10603 | - Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of. |
| MIL-G-10924 | - Grease, Automotive and Artillery. |
| MIL-R-12144 | - Reflector, Indicating, Clearance. |
| MIL-V-18436 | - Valves, Check: Bronze, Cast Iron, and Steel Body. |
| MIL-G-20241 | - Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive. |
| MIL-P-23377 | - Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant. |
| MIL-G-23827 | - Grease, Aircraft and Instrument, Gear and Actuator Screw. |



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| MIL-P-46093 | - Primer Coating, Synthetic (for Brake Drums). |
| MIL-A-46153 | - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package. |
| MIL-L-46167 | - Lubricating Oil, Internal Combustion Engine, Arctic. |
| MIL-S-81733 | - Sealing and Coating Compound, Corrosion Inhibitive. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-H28 | - Screw Thread Standards for Federal Services. |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-209 | - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment. |
| MIL-STD-461 | - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference. |
| MIL-STD-642 | - Identification Marking of Combat and Tactical Transport Vehicles. |
| MIL-STD-889 | - Dissimilar Metals. |
| MIL-STD-1400 | - Engines, Gasoline and Diesel, Methods of Test. |
| MIL-STD-1474 | - Noise Limits for Army Materiel. |
| MS33800 | - Connector, Receptacle, Electrical - Pin Contact, No. 12, 14 and 16 A.W.G., Waterproof. |
| MS35644 | - Filler Neck: Fuel Tank-Military Vehicles. |
| MS35645 | - Cap, Fuel Tank: Tactical Vehicle, Fording. |
| MS35829 | - Hinge, Butt, Continuous, (Piano) 0.075 Thickness, 0.187 and 0.250 Pin Dia. |
| MS35844 | - Plug, Machine Thread, Magnetic: Drain. |
| MS49006 | - Plug, Pipe, Magnetic-Headless; Aluminum, Iron or Steel. |
| MS51339 | - Coupler, Drawbar, Ring: Offset (Taper Shank), 60,000 Lb. GVW. |
| MS52125 | - Composite Light-Tail, Stop, Turn and Marker. |



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| MS53004 | - Valve, Relay, Emergency-Breakaway, Air Brake. |
| MS53063 | - Indicator, Air Cleaner, Intake, Restriction, Mechanical Type. |

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

AT

| | |
|---|---|
| C7070301 | - Chart, Electrical, Circuit, Numbers. |
| C7339458 | - Chain Assembly, Safety (Welded Construction). |
| D8347200 | - Schematic, Wiring Diagram, Cart-Type, Trailer, 24 Volts (Typical). |

DEPARTMENT OF TRANSPORTATION (DOT)

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents from a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 36 - Structural Steel.
ASTM A 48 - Gray Iron Castings.
ASTM A 269 - Seamless and Welded Austenitic Stainless Steel Tubing for General Service.
ASTM A 278 - Gray Iron Castings for Pressure-Containing Parts for Temperatures up to 650° F (345° C).
ASTM A 283 - Low and Intermediate Tensile Strength Carbon Steel Plates, Shapes, and Bars.
ASTM A 331 - Steel Bars, Alloy, Cold-Finished.
ASTM B 584 - Copper Alloy Sand Castings for General Applications.
ASTM D 3951 - Standard Practice for Commercial Packaging.



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(Application for copies should be addressed to American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE

Standards of the Hydraulic Institute.

(Application for copies should be addressed to the Hydraulic Institute, 122 East 42nd Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

TIRE AND RIM ASSOCIATION (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pumping unit shall consist of a centrifugal pump and a gasoline or diesel engine (see 3.11). The pumping unit shall be either skid mounted or wheel mounted (see 3.12).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.10).



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3.3.1 Material deterioration and control. The pumping unit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Environmental conditions.

3.4.1 Operating temperature. The pumping unit shall perform as specified herein in any ambient temperature from plus 32° F to plus 125° F.

3.4.2 Storage temperature. The pumping unit shall not be damaged by storage under the following conditions:

a. High temperature - plus 155° F for not less than 4 hours.

b. Low temperature - to minus 65° F for not less than 12 hours without benefit of solar radiation.

3.5 Safety. All shafts or rotating or moving parts of the pumping unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run into the engine, exhaust, or electrical equipment. Exhaust or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.



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3.6 Noise limits. The noise produced by the water pump shall not exceed MIL-STD-1474 paragraph 5.1.1.1 Category D (85dB(A), at the operators' position or other accasionally occupied positions, when tested in accordance with 4.5.2.9. If technical infeasibility is established via MIL-STD-1474 paragraph 5.1.1.2, or the means of reduction is too costly, the written permission of the procuring activity shall be requested to exceed the 85dB(A) limit. The precautions of MIL-STD-1474 (see 6.7) shall be provided if and only if MIL-STD-1474 paragraph 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity. Hazard signs shall conform with MIL-STD-1474 paragraph 4.3 and be readable at the distance indicated on the sign. Hearing protectors shall not be supplied with the pump units.

3.7 Reliability. The specified mean-time-between-failure (MTBF) shall be 100 hours when the pumping unit is tested as specified in 4.5.2.10.

3.8 Maintainability.

3.8.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.5 and 6.6), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.11.

3.8.2 Maintenance ratio. The pumping unit shall have a maintenance ratio of not more than 0.10 for gasoline-engine-driven and not more than 0.08 for diesel-engine-driven pumps. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.9 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.10 Pumps. The pump shall be of the single-stage single-suction centrifugal type. Each pump shall be provided with replaceable wearing surfaces and a pump case drain port and plug. The drain port shall be internally threaded with American standard taper pipe threads (NPT) and shall be of such size and located so as to drain the pump case in not more than 2 minutes. The drainplug shall be a square-head pipe plug conforming to WW-P-471, type A, class A, material B, and finish U. When the drainplug is in the volute passage, the plug shall be flush with the inner surface of the pump case. The direction of suction of each pump shall be horizontal. The direction of discharge of each pump shall be horizontal either by the design of the pump case or by the use of fittings.



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3.10.1 Hydrostatic pressure. The pump shall not leak or show permanent deformation when subjected to an internal hydrostatic pressure equal to 1-1/2 times the shutoff head developed at full throttle operation.

3.10.2 Pump case. The pump case shall be an iron casting conforming to ASTM A 278, class 20. The pump case shall be provided with replaceable wearing rings, or wearing plates. The wearing rings shall be bronze conforming to ASTM B 584, alloy C83600. The wearing plates shall be steel conforming to QQ-S-635, No. 1020.

3.10.3 Pump impeller. Unless otherwise specified herein, the impeller shall be an iron casting conforming to ASTM A 278, class 20, and shall be the semi-open vane nonclogging type. The impeller shall be keyed to the impeller shaft and held in position by a lock-cap nut with a shaft-protecting washer, or the impeller shall be screwed on the shaft in a direction opposite to the direction of rotation with enough piloting surface to insure that the impeller is normal to the axis of the shaft.

3.10.3.1 Impeller balancing. The impeller, mounted on its shaft, shall be dynamically balanced. The out-of-balance condition shall be not more than 0.05 ounce-inch at the rated operating speed of the pump.

3.10.4 Pump shaft sleeve. A shaft sleeve shall be provided between the pump shaft and the seal. The shaft sleeve shall be aluminum-bronze or corrosion-resistant steel conforming to ASTM A 269, grade optional, as specified (see 6.2).

3.10.5 Pump shaft seal. The pump shall be provided with a mechanical-type self-lubricating seal to prevent leakage. The seal shall be self-alining and shall be removable from the pump shaft without disassembling the pump from the engine mounting face. The adjustment of the sealing surface shall be independent of the impeller adjustment. The seal may be fabricated from dissimilar non-corrosive materials of different hardness properties. All components of the mechanical-type seal shall be resistant to corrosion.

3.10.6 Priming. When operating at conditions equivalent to sea level atmospheric conditions of 29.92 inches of mercury, and with water with a specific weight of 62.3 pounds per cubic foot at a temperature of 68° F, the pump shall perform as follows:

a. Prime in not more than 3 minutes when subjected to a static suction lift equivalent to 25 feet, using 30 feet of appropriate size suction line and one elbow.

b. Prime in not more than 1 minute when subjected to a static suction lift equivalent to 10 feet, using 15 feet of appropriate size suction line and one elbow.



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The suction and discharge fittings shall be the same nominal size as the suction and discharge ports of the pump.

3.10.7 Check valve. A check valve shall be provided on the suction inlet of each pump. When subjected to a static discharge head equal to the maximum total head indicated on figures 1 through 4, as applicable, the valve shall prevent siphoning of water from the pump case and shall not leak or show permanent deformation of the body, cover, seat, or any of its components.

3.11 Engine. The pumping unit shall be powered by a gasoline or diesel engine, as specified (see 6.2).

3.11.1 Gasoline. The gasoline engine, including all systems, components, accessories, and auxiliaries, shall be a heavy-duty 4-cycle standard commercial-industrial type capable of operating on gasoline conforming to VV-G-1690 and MIL-G-3056 and lubricating oil as specified herein. If the normal operating oil pressure of the engine is 15 pounds per square inch (psi) or above, a full-flow oil filter shall be furnished. A dry-type air cleaner shall be furnished. The engine, except for the type I, size 6 pump, shall be air cooled. The type I, size 6 pump may be air or liquid cooled. The temperature of the coolant in the radiator top tank (if liquid cooled) shall not exceed 210° F when the engine is operated under any condition specified herein. The engine shall have both manual and electrical starting systems. The electrical battery starting and charging system shall be 12-volt negative ground. The engine shall be furnished with a housing. When wheel mounting is specified, the engine radiator or cooling air inlet shall have a latching hinged door. The door shall be marked in bold lettering: "OPEN WHEN OPERATING". The engine shall be the same engine that has been used in the manufacturer's standard commercial pump of the same size and capacity for at least 1 year.

3.11.2 Diesel. The diesel engine, including all systems, components, accessories, and auxiliaries, shall be a heavy-duty standard commercial-industrial type capable of operating on diesel fuel conforming to VV-F-800 and lubricating oil as specified herein. If the normal operating oil pressure of the engine is 15 psi or above, a full-flow oil filter shall be furnished. A dry-type air cleaner shall be furnished. The engine, except for the type I, size 6 pump shall be air cooled. The type I, size 6 pump may be either air or liquid cooled. The temperature of the oil in the oil sump shall not exceed 250° F and the temperature of the coolant in the radiator top tank, if liquid cooled, shall not exceed 210° F when the engine is operated under any conditions specified herein. The engine shall be equipped with a 12-volt negative ground electrical starting and charging system. The engine shall be equipped with a housing. When wheel mounting is specified, the engine radiator or cooling air inlet shall have a latching hinged door. The door shall be marked in bold lettering: "OPEN WHEN OPERATING". The engine shall be the same engine which has been used in the manufacturer's standard commercial pump of the same size and capacity for at least 1 year.



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3.11.3 Power and speed rating. The power and speed rating of the gasoline or diesel engine shall be such that operation of the pump under any of the operating conditions specified herein will not require horsepower in excess of the continuous rating of the engine as previously published by the engine manufacturer, or 85 percent of the engine manufacturer's previously published maximum horsepower rating at the applied speed, whichever is less.

3.11.4 Filling, draining, and checking provisions. The engine crank-case shall be equipped with a dipstick to determine the fluid levels. The drain outlet shall be filled with a magnetic drainplug having either tapered pipe threads or shoulder and gasket, size conforming to MS35844 or MS49006. In lieu of a magnetic drainplug, a separate magnetic device may be fastened inside the enclosure at or near the bottom and shall be removable for cleaning without removal of other parts or components. Each drainplug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure. When the pumping unit is in a level position, drainage of fluids shall be to a container without draining on or over any part of the pumping unit. Integral tubes, hose, or troughs may be used to convey the fluids from the drain outlet. Accessibility to the fill opening, the fluid level checking device, and the drainplug or valve shall be provided without removal or adjustment of accessories and parts except plates equipped with hand-operable quick-disconnect fasteners or capscrews in applications where quick-disconnect fasteners are subject to damage. The drain outlets for the engine cooling system of liquid-cooled engines shall provide complete drainage of the system. The radiator drain shall be such that coolant will not drain onto any part of the pumping unit. An extension hose shall be permitted.

3.11.5 Instrumentation. An instrument panel shall be furnished and shall be equipped with the following instruments:

a. Lubricating oil pressure gage.

b. Tachometer.

c. Ammeter, battery charging.

d. Hourmeter.

e. Cooling liquid temperature indicator.

f. Air cleaner restriction indicator conforming to MS53063.

g. Fuel pressure gage (diesel).

The tachometer and hourmeter may be incorporated into one instrument provided the hourmeter corresponds to the clock time at the normal pump operating speed.



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3.11.6 Control device system. The liquid-cooled engines shall be provided with a high cooling liquid temperature device for stopping the engine if the engine reaches an overheated condition. The air-cooled engines shall be provided with a high cylinder head temperature device for stopping the engine if the engine reaches an overheated condition. Both liquid- and air-cooled engines shall be provided with a low lubricating oil pressure control device for stopping the engine when a low lubricating oil pressure condition exists. Each safety device shall have individual signal lights and manual reset (if reset is required) mounted on the instrument panel. The signals shall indicate which condition has caused the shutdown and shall be marked as to their function. All control devices shall be designed to actuate at the engine manufacturer's recommended temperatures and pressures.

3.11.7 Manual engine controls. Manual engine controls for gasoline engines shall include an ignition switch to activate the entire electrical circuit, hand choke, and throttle. Manual engine controls for diesel engines shall include a switch to activate the engine electrical circuits, including fuel solenoid valve, safety control devices, and starter and battery charging circuits. A waterproof push button starter switch shall be furnished. The throttle shall be the micrometer type and shall permit operation of the engine at any selected governed speed from idling to maximum governed speed. The throttle shall be provided with a convenient and positive means for locking in any position.

3.11.8 Fuel tank. A fuel tank shall be furnished for both gasoline and diesel engines. The capacity of the tank shall be sufficient for not less than 4 hours of continuous operation at the rated load. The fuel tank shall be equipped with a cap conforming to MS35645, a strainer, and a neck conforming to MS35644. The tank shall be provided with a hand-operated fuel shutoff valve attached directly to the tank and a mechanically actuated fuel gage mounted on the tank. The fuel tank outlet shall be not less than 1/2 inch above the bottom of the tank. The tank outlet or the fuel line shall be provided with a 3-way selector valve and a 1/2-inch 20 SAE flared fitting for connection with an auxiliary fuel supply. A corrosion-resisting circular metal face plate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK", and "AUXILIARY".

3.11.9 Battery-disconnect switch. The pump unit shall be provided with a battery-disconnect switch to permit disconnecting the battery in the event of an emergency. The switch shall be installed in a convenient position on the instrument panel and shall withstand the full cranking motor load.

3.12 Mounting. The pumping unit shall be skid mounted or wheel mounted, as specified (see 6.2).



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3.12.1 Skid mounted. The skid base shall be structural steel conforming to ASTM A 283, grade D. The skid base shall maintain alinement when the pumping unit is being towed and handled in the field. The skid shall be provided with upturned ends and shall be equipped with a retractable towing eye at each end of the skid base. The retractable towing eyes shall be located so that the possibility of overturning the pumping unit is minimized and positive tracking is assured when the unit is being towed. Each towing eye shall be a U-bolt with an opening of not less than 2-1/2 inches in diameter when extended.

3.12.2 Wheel mounted. The pumping unit shall be mounted on two wheels, as specified herein.

3.12.2.1 Gross vehicle weight. The gross vehicle weight (GVW) of the 2-wheel-mounted pumping unit shall be not more than the following:

a. 2000 pounds for type I, size 4, classes A and B.

b. 3500 pounds for type I, size 6 and type II, size 4.

3.12.2.2 Rated payload. The rated payload of the 2-wheel-mounted pumping unit for unrestricted cross-country operations shall be not less than the following:

a. 1000 pounds for type I, size 4, classes A and B.

b. 2500 pounds for type I, size 6 and type II, size 4.

3.12.2.3 Lunette loads. The load on the lunette shall be as follows:

a. The maximum load shall be not more than 200 pounds and the minimum load shall be not less than 100 pounds for type I, size 4, classes A and B.

b. The maximum load shall be not more than 350 pounds and the minimum load shall be not less than 150 pounds for type I, size 6 and type II, size 4.

3.12.2.4 Dimensions. The 2-wheel-mounted pumping unit shall conform to the following limitations:

| | |
|---|---|
| Overall length, maximum | - 100 inches |
| Overall width, maximum | - 75 inches |
| Overall height, maximum | - 70 inches |
| Overall clearance, minimum | - 14 inches |
| Tread, axle, minimum | - 62 inches |



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3.12.2.5 Mobility characteristics. When towed by a military cargo truck of the proper size (see 3.21), the pumping unit shall perform without malfunction, damage, or permanent deformation to any of its components when subjected to the following conditions:

a. Cross-country operation. Over open, rolling, hilly, or cross-country terrain at speeds up to 20 miles per hour (mph).

b. Unimproved road operation. Over unimproved and secondary roads at speeds up to 30 mph.

c. Highway operation. Over smooth hard-surfaced roads at speeds up to 50 mph.

3.12.2.6 Trailing and turning ability. When coupled to a military cargo truck of the proper size (see 3.21), the pumping unit shall track without bottoming, weaving, or sideswaying. The turning clearance of the pumping unit shall be such that, when the pumping unit becomes jackknifed, its longitudinal centerline shall form a minimum horizontal angle of 40 degrees to the right and 40 degrees to the left with the longitudinal centerline of the truck. Turns shall be performed without interference or damage to either the pumping unit or the truck at the truck's minimum turning radius.

3.12.2.7 Frame. The frame shall be structural steel conforming to ASTM A 36 and shall be suitable for the installation of the pump, engine, and running gear.

3.12.2.8 Lunette and support legs. The pumping unit shall be equipped with a lunette and three support legs.

3.12.2.8.1 Lunette. The lunette shall conform to MS51339 and shall be installed so that the offset of the eye may be reversed.

3.12.2.8.2 Front support leg (lunette end). A front support leg and bracket assembly shall be provided and shall be installed so that the level ground height of the lunette shall be 15 inches, plus or minus 1/2 inch, when rotated to the lower position. The leg shall fold to provide ground clearance and not reduce the angle of approach.

3.12.2.8.3 Rear support legs. Two rear support legs shall be provided to prevent the frame from tipping over when hose or pipe is coupled to the pump or when the pump or bearing housing is being filled or serviced by personnel. The legs shall be capable of being securely locked in both the retracted and extended positions. The locking device and the legs shall be captive-chained to the pump unit frame to prevent loss. The legs shall be capable of being adjusted in 2-inch increments from the fully retracted position to the fully extended position (and vice versa) by hand. When in the fully retracted position, the jack stand shall not protrude more than 2 inches below the bottom of the side member of the frame.



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3.12.2.9 Intervehicular connectors. Intervehicular connectors shall be installed on the pumping unit so that they shall be extended between 36 and 42 inches toward the prime mover from the center of the lunette eye. A clip, in accordance with SAE J702, shall be furnished to securely hold the electrical harness plug and to double the harness back on the tongue or frame. These connectors shall not interfere with vehicle operations when tested in accordance with 4.5.2.6.2.

3.12.2.10 Safety chains. The pumping unit shall be equipped with two safety chains conforming to Drawing C7339458, except that the length of each chain shall be the same as the intervehicular connectors. The forged eyes of the safety chains shall be perimeter welded or bolted to the tongue or frame.

3.12.2.11 Axle assembly. The axle assembly shall include spindle, hub, backing plate, brakes, and wheel components, as specified herein. The rated capacity of the axle shall be not less than the following:

a. 3000 pounds for type I, size 4, classes A and B.

b. 5000 pounds for type I, size 6 and type II, size 4.

3.12.2.11.1 Suspension. The axle suspension system shall be of a springing capacity to sustain the rated payload under the performance characteristics specified herein without transmitting damaging shock to the pumping unit. The suspension shall result in as low a center of gravity as possible without affecting the ground clearance requirements. Rubber bumper stops shall be provided to prevent metal-to-metal bottoming of the suspension.

3.12.2.11.2 Shock absorbers. Two telescoping-type shock absorbers without linkage shall be mounted on the frame and shall control jounce and rebound.

3.12.2.11.3 Wheels. Wheels shall be provided with ball or roller bearings and shall be demountable at the hub. The weight of the pumping unit shall be proportionately distributed over the wheels so as to assure proper balance. Rims shall comply with the recommendations of the TRA Yearbook.

3.12.2.11.4 Tires. At the option of the contractor, tires shall be tube type, tubeless type with nondirectional mud and snow tread (NDMS), nondirectional cross-country tread (NDCC), or open diamond tread which will not pick up stones or foreign objects. Tires shall be rated to support the gross weight of the pumping unit without exceeding 45 psi internal air pressure. When tube-type tires are furnished, inner tubes shall be heavy-duty type and shall be of proper size for the tires.

3.12.2.11.5 Wheel assembly balancing. Each assembly of tire, tube, wheel, and hub components shall be balanced to preclude wheel tramp and drumming at road speeds up to 50 mph.



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3.12.2.12 Brake system. The 2-wheel-mounted pumping unit shall be equipped with an air-over-hydraulic brake system. All components and lines of the system shall be installed and protected so as to prevent damage or malfunction from water, ground obstructions, and debris. The air-over-hydraulic power cluster(s) shall be the tandem master cylinder type or, at the contractor's option, shall consist of two power clusters, each having a single master cylinder. A rupture or leakage-type failure of any single hydraulic system pressure component shall not prevent actuation of the other system and the actuated brakes shall not cause unbalanced braking or swerving of the pumping unit. Not less than 5 inches of slack in the flexible air lines shall be provided at each of the emergency relay valve connections to permit interchange of the various makes of MS53004 valves. All brake lines shall be installed to facilitate replacement from bulk supply. All nonlubricated metal surfaces that are enclosed by the brake drum and the backing plate shall be coated with primer conforming to MIL-P-46093. Moving parts shall be coated with grease conforming to MIL-G-10924.

3.12.2.12.1 Braking ability. The combined service brakes of the assembled towing and towed vehicle shall control, decelerate, and stop the assembly in conformance to DOT Motor Carrier Safety Regulations, except that the stopping distance shall be accurately measured from the point of initial application of the brake pedal to the stopping point. When actuated, the emergency breakaway system shall produce emergency brake application in accordance with DOT Motor Carrier Safety Regulations.

3.12.2.12.2 Parking brake. The pumping unit shall be equipped with a parking brake with positive locking device. The parking brake shall be capable of holding the pumping unit on a 20-percent-grade dry paved ramp, while heading either up or down ramp, without evidence of slippage.

3.12.2.13 Vehicular electrical system. The pumping unit shall be equipped with a 24-volt complete vehicular electrical system conforming to Drawing D8347200, and as specified herein. All wiring and all components shall be located or guarded to provide protection against damage and breakage. The friction retainment wiring connectors shall conform to MS33800 and shall be lubricated prior to assembly with paste-type electrical insulation compound. Each connector assembly shall be secured by the clip shown on Drawing D8347200. All wires shall be number coded near each connection in accordance with the wiring diagram and Drawing C7070301. Wires and harnesses shall be fastened by the support strap at intervals of not more than 3 feet and shall be routed within the framing where possible. Except for the intervehicular harness, unsupported wires are not acceptable. Grommets shall be installed where wires or harnesses pass through sheet metal and framing members.



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3.12.2.14 Lighting. The pumping unit shall be equipped with a service and blackout lighting system. The service lighting shall conform to the requirements of DOT Motor Carrier Safety Regulations. Two taillamps conforming to MS52125 shall be installed at the rear of the pumping unit as far apart as possible to indicate vehicle width and for turn-signal operation. A blackout stoplight conforming to MS52125 shall be installed to the left of the right taillamp.

3.12.2.15 Reflectors. Reflectors conforming to MIL-R-12144, types I and II, shall be furnished and shall be mounted in accordance with DOT Motor Carrier Safety Regulations. In addition, one amber reflector shall be mounted facing forward on each front corner and one red reflector shall be mounted on each side near the rear of the pumping unit and shall be installed at approximately the same height as those installed on the rear. All reflector mounting locations shall give the maximum reflected warning.

3.13 Electromagnetic interference. The electromagnetic interference emission characteristics of the pumping unit shall conform to MIL-STD-461, class C1.

3.14 Lubrication.

3.14.1 Lubricants. The pumping unit shall operate as specified herein when serviced with the following military lubricants:

Engine: MIL-L-2104 or MIL-L-46167.

Axle: MIL-L-2104, MIL-L-2105, or MIL-L-46167.

Hydraulic system: MIL-L-2104 or MIL-L-46167.

Grease lubrication: MIL-G-10924 or MIL-G-23827.

3.14.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable quick-disconnect fasteners may be removed to provide accessibility.

3.14.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.15 Slinging provisions. Permanently affixed slinging provisions that enable the pumping unit to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209.



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3.16 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the pumping unit to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209.

3.17 Identification marking. The pumping unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pumping unit on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.17.1 Identification marking of pump inlet and outlet. The pump suction and discharge shall be identified by markings integrally cast in the pump or by plates specified in 3.17 permanently secured in place by means of metal fasteners.

3.17.2 Wheel mount marking. Marking of the wheel-mounted pumping unit shall be in accordance with MIL-STD-642. The gross vehicle weight shall be so marked on each side of the unit in such a manner as to be readily discernible. The prescribed tire pressure shall be marked on the side near the tires.

3.17.3 Skid mount marking. The gross weight shall be stenciled near the center on both sides of each skid-mounted pumping unit. The letters shall be Gothic capitals and the numerals shall be Arabic. The letters and numerals shall be 2 inches high where space permits.

3.18 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material shall be provided. In addition, the plate shall show the silhouette of the pumping unit in transport position, indicating the center of gravity and the location and capacity of the slinging and tiedown provisions. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.19 Treatment and painting. The portions of the pumping unit normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.20 Toolbox. A toolbox made of steel not less than 0.071 inch in thickness shall be attached to the pumping unit. The toolbox shall contain all the tools required for the general maintenance and operation of the pumping unit. The toolbox lid shall be secured to one edge of a side by a continuous hinge conforming to MS35829 extending the full length of the box edge. The lid shall be held in the closed position by not less than two fasteners. The hasp shall be suitably located to secure the lid to the box and to accommodate a lock. The toolbox shall be securely mounted in an accessible protected location permitting the lid to be opened through an arc of 90 degrees. The lid shall be held fully open by the weight of the lid or a suitable fastener. Adequate drainage of the toolbox shall be provided.



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3.21 Government-loaned property. When specified (see 6.2), one truck, cargo, of the current design and suitable size military vehicle shall be loaned by the Government (see 6.4).

3.22 Rail. When specified (see 6.2), the unpackaged pumping unit shall be subjected to the rail humping test specified in 4.5.2.12. Upon completion of the test, the pumping unit shall show no evidence of damage or permanent deformation to any part or component. After testing, the pumping unit shall operate as specified herein for a period of not less than 1 hour without malfunction or permanent deformation of any part of the pumping unit.

3.23 Type I. The type I pumping unit, once the case has been filled with water, shall be the integral self-priming type. The pump shall have a peak efficiency of not less than 60 percent.

3.23.1 Size 4-inch suction and discharge pumping unit. The size 4 pumping unit shall have 4-inch nominal pipe size suction and discharge ports. The suction and discharge ports shall be internally threaded NPT.

3.23.1.1 Class A pumping unit. The class A pumping unit shall deliver the capacities shown on figure 1. When operating against a total head of 30 feet and using a 25-foot static suction lift, the pumping unit shall deliver not less than 250 gallons per minute (gpm).

3.23.1.2 Class B pumping unit. The class B pumping unit shall deliver the capacities shown on figure 2. When operating against a total head of 30 feet and using a 25-foot static suction lift, the pump shall deliver not less than 350 gpm.

3.23.2 Size 6-inch suction and discharge pumping unit. The size 6 pumping unit shall have 6-inch nominal pipe size suction and discharge ports. The suction and discharge ports shall have cast-iron flanges conforming to WW-F-406, class C-125 pounds. The capacity of size 6 pumps shall be as shown in figure 3.

3.23.2.1 Check valve. Unless otherwise specified (see 6.2), a check valve shall be installed in the suction end of each size 6 pumping unit. The check valve shall conform to MIL-V-18436, group B, type I, style A, class 250.

3.23.2.2 Impeller shaft. The impeller shaft shall be steel alloy conforming to ASTM A 331, grade 4140. The shaft shall have an ultimate tensile strength of not less than 130,000 psi.



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3.23.2.3 Coupling. A flexible or spline-type coupling shall be provided which shall enable the pump and bearing housing to be removed from the engine as a unit by uncoupling the drive half from the driven half of the coupling. The coupling shall be the dry type and shall provide for misalinement and end float. The coupling shall transmit power from the engine to the impeller shaft through an arrangement which will reduce transmission of torsional vibration. The coupling shall have a horsepower rating in excess of the maximum brake horsepower required by the pump.

3.23.2.4 Intermediate bearing housing. Close-coupling shall be accomplished through the use of an intermediate bearing housing. The housing shall mount directly to the engine flywheel housing and to the pump case. The housing shall be provided with antifriction bearings of such size and type so as to absorb the maximum radial and thrust loads imposed by the pump impeller. The bearings shall be either sealed for life or in a grease-filled enclosure that will not require periodic refilling except when the bearings are removed from the enclosure. The housing shall be cast iron conforming to ASTM A 48, class 30.

3.24 Type II, size 4. The type II, size 4 pumping unit shall be the vertical split-casting auxiliary self-priming type, with closed impeller and a horizontal discharge port. The pump shall be designed so that the pump seal may be subjected to either atmospheric pressure or below atmospheric pressure under the specified performance conditions. The suction and discharge ports of each pump shall have flanges and threaded companion flanges in accordance with WW-F-406, class optional. The pump case shall be provided with openings and fittings for venting and priming. Each pumping unit shall be provided with replaceable wearing rings.

3.24.1 Performance. Each pumping unit shall continuously deliver the capacities shown on figure 4, curve A. The pumping unit shall also deliver intermittent capacities as shown on figure 4, curve B. The pumping unit shall have a peak efficiency of not less than 70 percent. The pumping unit efficiency at any point on the pump efficiency curve shall permit the pump capacities to be increased from the continuous performance to the intermittent performance, as shown on figure 4, curves A and B. A check valve shall be provided on the discharge side of the pump.

3.24.2 Pump primer. The pump primer shall be connected to the pump case at the highest point on the suction nozzle. Air shall not be trapped anywhere within the primer. The primer case shall be provided with tapped openings and fittings for venting and for rapid and complete draining.



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3.24.3 Pump impeller. The pump impeller shall be an iron casting conforming to ASTM A 278, class 40, and shall be single-suction close-type operating at optimum distance from the volute tongue and the sides of the pump case and cover walls.

3.24.4 Impeller shaft. The impeller shaft shall be as specified in 3.23.2.2.

3.24.5 Coupling. The coupling shall be as specified in 3.23.2.3.

3.24.6 Bearing housing. The bearing housing shall be as specified in 3.23.2.4.

3.25 Threads. Screw threads shall be in accordance with FED-STD-H28.

3.26 Workmanship. All parts, components, and assemblies of the pumping unit, including castings, forgings, molded parts, stampings, bearing seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or beveled.

3.26.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.26.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.



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4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article pumping unit shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article pumping unit shall be tested as specified in 4.5.2 through 4.5.2.12. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for the minor defect.

4.4.3 Tests.

4.4.3.1 Individual. Each pumping unit shall be subjected to the tests specified in 4.5.2.2.1 and 4.5.2.2.2.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.5 through 4.5.2.8.

4.5 Inspection procedure.

4.5.1 Examination. The pumping units shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.



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103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Safety not as specified.
106. Major assemblies, installed attachments, and drain outlets not accessible for maintenance, repair, or replacement.
107. Removable covers and plates not equipped with quick-disconnect fasteners.
108. Pump not as specified.
109. Pump fittings not the same size as the pump suction and discharge ports.
110. Pump impeller not as specified.
111. Flexible or spline type coupling not as specified.
112. Suction and discharge ports not internally threaded for size 4 unit.
113. Bearing housing not as specified.
114. Pump shaft sleeve not as specified.
115. Adjustment of pump shaft seal not independent of the adjustment of the impeller shaft.
116. Filling, draining, and checking of engine crankcase not as specified.
117. Instrumentation of the engine not as specified.
118. Manual engine controls not as specified.
119. Engine safety devices missing or not as specified.
120. Doors for cooling air intake on wheel mounted unit missing or not as specified, or sign on doors missing.
121. Fuel tank not as specified.
122. Engine not equipped with 3-way fuel valve.
123. Battery disconnect switch missing.
124. Skids on skid mounted unit not as specified.
125. Wheel mounted unit not as specified.
126. Dimensions of wheel mounted unit not within tolerance specified.
127. Frame not suitable for installation of pump, engine, and running gear.
128. Lunette missing.
129. Front or rear support legs missing or not adjustable.
130. Safety chains missing.
131. Rated capacity of axle less than specified.
132. Rubber bumper stops not provided for axle suspension.
133. Shock absorbers not as specified.
134. Parking brake positive locking device not provided.
135. Vehicular electrical wiring and lighting system not as specified on wheel mounted unit.
136. Reflectors not as specified or not mounted as specified on wheel mounted unit.
137. Lubrication fittings not as specified or missing.



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138. Pressure release device missing.
139. Identification or instruction plates missing or not as specified.
140. Tool box not as specified.
141. Screw threads not as specified.
142. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.

4.5.2 Tests. Prior to any test, the complete pumping unit shall be lubricated and serviced. Lubrication shall be with the lubricants specified in 3.14. Servicing shall be in accordance with the contractor's recommended practice.

4.5.2.1 Engine (gasoline and diesel). Prior to examination and tests of the pumping unit, the engine which will be installed in the first article units shall be tested in accordance with test method 2000 of MIL-STD-1400, except that test method 2400 shall not apply. One engine out of each lot of 50 or less shall be tested in accordance with test method 3000, MIL-STD-1400. Each engine furnished, except the engines selected for the first article pumping units and production control tests (test method 3000), shall be tested in accordance with test method series 4000, MIL-STD-1400, except that the engine manufacturer's standard commercial run-in may be used in lieu of test method series 4000 provided it equals or exceeds the requirements of test method series 4000, MIL-STD-1400. Failure to meet any requirements in the above tests shall constitute failure of this test.

4.5.2.1.1 Engine power (gasoline). During conduct of the pumping unit performance tests specified in 4.5.2.5, the engine rpm and intake manifold vacuum shall be recorded at not less than 30-minute intervals. These readings shall be compared with the engine speed (rpm) brake horsepower, and intake manifold vacuum curve for the continuous brake horsepower conditions as established during conduct of test method series 2000 of MIL-STD-1400. Any intake manifold vacuum reading recorded during the pump performance tests that is lower than the reading listed on the continuous horsepower curve established during conduct of test method 2300 of MIL-STD-1400, at the corresponding speed (rpm), shall constitute failure of this test.

4.5.2.2 Pump preassembly.

4.5.2.2.1 Impeller. Prior to assembly, the pump impeller shall be tested for out-of-balance by a dynamometer at the operating speed of the pump. An out-of-balance condition of more than 1.0 ounce-inch shall constitute failure of this test.

4.5.2.2.2 Hydrostatic. Subject the pump to a hydrostatic pressure equal to 1-1/2 times the shutoff head obtained at full throttle operation. Leakage or permanent deformation of the pump shall constitute failure of this test.



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4.5.2.2.3 Brake horsepower. Prior to assembling the pump to the engine, the brake horsepower required to operate the pump at rated capacity shall be determined. Nonconformance to 3.11.3 shall constitute failure of this test.

4.5.2.3 Electromagnetic interference. The pumping units shall be tested to determine conformance to 3.13. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.4 Storage and operational temperature. Place the pumping unit in a shelter or container equipped for heating and cooling. Stabilize temperature at 155° F for not less than 4 hours. Lower and stabilize temperature at 125° F and operate the pump for not less than 4 hours. Lower and stabilize temperature at 32° F and operate pump for not less than 4 hours. Stabilize shelter or container at minus 65° F for 12 hours without benefit of solar radiation. Raise and stabilize temperature at 32° F and operate pump for not less than 4 hours. Stabilize temperature at 125° F and operate pump for not less than 4 hours. Nonconformance to 3.4.1 or 3.4.2 shall constitute failure of this test.

4.5.2.5 Performance. The performance test of the pumping unit shall be conducted in accordance with the applicable provisions of the test code for centrifugal pumps of the Standards of the Hydraulic Institute. All data shall be corrected to standard sea level atmospheric conditions of 29.92 inches of mercury and to the specific weight of water of 62.3 pounds per cubic foot at a temperature of 68° F.

4.5.2.5.1 Priming. The pumping unit shall be primed and operated under the conditions specified herein. Nonconformance to 3.10.6 shall constitute failure of this test.

4.5.2.5.2 Capacity. The pumping unit shall be primed and operated to determine the head capacities shown in figure 1 through 4, as applicable. Nonconformance to 3.23.1.1, 3.23.1.2, 3.23.2, or 3.24.1, as applicable, shall constitute failure of this test.

4.5.2.5.3 Check valve. After the pump has been completely primed, subject the check valve to a maximum test pressure (total static discharge head) equal to the maximum total head as shown on the applicable figure. Shut off the pump and allow it to sit for not less than 30 minutes. Restart the pump. Determine whether the pump primes itself. Self-priming of the pump is an indication that the valve prevents siphoning of water from the pump case. Examine the check valve. Failure of the pump to prime itself, leakage, or permanent deformation of any component of the valve shall constitute failure of this test.



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4.5.2.5.4 Pump case drain. With the pump case filled with water, remove the drainplug. Inability of the drain to remove all the water from the case in 2 minutes shall constitute failure of this test.

4.5.2.6 Mobility.

4.5.2.6.1 Skid tow. The pumping unit, mounted on skids, shall be towed over a concrete surface for a distance of not less than 100 feet. Evidence of damage to or permanent deformation of the skid mounting shall constitute failure of this test.

4.5.2.6.2 Trailer towing. The pumping unit, mounted on its trailer, shall be coupled to the Government-loaned truck (see 3.21) and shall be towed as follows:

a. Cross-country - 50 miles at 20 mph.

b. Unimproved road - 150 miles at 30 mph.

c. Paved highway - 200 miles at 50 mph.

After this portion of the test, perform 10 left-circle and 10 right-circle turns at minimum radius on surfaces specified in a, b, and c above. Nonconformance to 3.12.2.5 or 3.12.2.6 shall constitute failure of this test.

4.5.2.7 Tiedown provisions.

4.5.2.7.1 Test procedure. With the pumping unit anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns.

4.5.2.7.2 Failure criteria. Weld failure, permanent deformation of the pumping unit or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.8 Slinging provisions.

4.5.2.8.1 Test procedure.

a. Lift the pumping unit and hold it in suspension in its normal travel position, using slings that converge not more than 24 feet above the lowest extremity of the pumping unit. Determine the forward angles of application for each provision. Measure the clearance between each sling and the pumping unit. Measure the eye openings and clearance dimensions of the provisions.



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b. Restrain the pumping unit by anchoring the main frame and subject each slinging provision to a force of 2-1/2 times the load it carried when suspended. This force shall be applied in the direction as determined in a. above. An acceptable alternate method of test shall be to lift the pumping unit as in a. above and add weights to the pumping unit until the force on each provision is 2-1/2 times the initial force. Hold the pumping unit under either of the above loading conditions for not less than 90 seconds.

4.5.2.8.2 Failure criteria. Weld failure, permanent deformation of the pumping unit or any slinging provision, slings converging more than 24 feet above the lowest extremity of the pumping unit, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.9 Noise level measurement. Noise levels shall be measured in accordance with MIL-STD-1474, for steady-state noise. As a minimum: Noise levels shall be measured when equipment is operating at normal maximum rated load; Measurements will be taken at the noisiest location, operatiors location, and at a 25-foot radius from the geometric center of the equipment every 45 degree ±5 degree of azimuth, 65±1 inches above the ground. Should the noise level exceed 85dB(A) at any noise test point, failure to comply with paragraph 3.6 or MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.10 Reliability. Using the MTBF specified in 3.7, the pumping unit shall be tested for reliability. With the pump operating at the conditions of maximum water horsepower indicated on the appropriate performance curve (see figure 1 through 4), test until an "accept" or "reject" decision is reached, in accordance with figure 5. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair, or replacement action using controls and on-equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.

b. Damage to the pumping unit by continued operation.

c. Safety hazards to personnel.

Nonconformance to 3.7, 3.10.1, 3.10.3.1, 3.10.5, 3.10.6, or 3.10.7 shall constitute failure of this test.



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4.5.2.11 Maintainability.

4.5.2.11.1 Maintenance support evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.8.1. Failure of the support elements to maintain the pumping unit as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTFB.

4.5.2.11.2 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.8.2 shall constitute failure of this test.

4.5.2.12 Railroad car impact. Anchor the unpackaged pumping unit on a domestic service railway freight car (blocking is permitted). Hump the car, with the pumping unit tied down, at speeds of not less than 4, 6, and 8 mph, respectively, against five empty freight cars with their brakes set and draft gear extended. The pumping unit shall face in the direction of the five cars. Repeat the procedure at not less than 8 mph with the rear of the pumping unit in the direction of the five cars. The pumping unit may be reversed by either reversing the direction of the flatcar (preferred) or by reloading the pumping unit in the proper direction on the freight car. Once the test has begun, there shall be no readjustment of load nor any reconditioning of the blocks, chocking material, or tiedowns except as necessary to reverse the direction of the pumping unit. After each impact, the pumping unit shall be inspected. After completion of the railroad impact test, an operational checkout of the pumping unit shall be conducted. Nonconformance to 3.22 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking of the pumps shall be examined and tested in accordance with the quality assurance provisions of MIL-P-10603.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B or commercial as specified (see 6.2).

5.1.1 Level A. Each complete pump shall be preserved in accordance with the level A requirements of MIL-P-10603 except the cooling system of liquid cooled engines shall be preserved by the water and antifreeze procedure specified in MIL-E-10062, using antifreeze conforming to MIL-A-46153.

5.1.2 Level B. Each complete pump shall be preserved in accordance with the level B requirements of MIL-P-10603 except the cooling system shall be preserved as specified in 5.1.1.



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5.1.3 Commercial. Each complete pump shall be preserved in accordance with ASTM D 3951 except the engine shall be preserved in accordance with MIL-E-10062, level C, as specified for alternate preservation and packaging of liquid cooling systems shall be preserved as specified in 5.1.1.

5.2 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).

5.2.1 Level A. Each complete pump, preserved as specified in 5.1, shall be packed in accordance with the level A requirements of MIL-P-10603.

5.2.2 Level B. Each complete pump, preserved as specified in 5.1, shall be packed in accordance with the level B requirements of MIL-P-10603.

5.2.3 Commercial. Each complete pump, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. Additionally, each shipping container shall be marked with cube and gross weight.

6. NOTES

6.1 Intended use. The pumping units are intended for use in general purpose water pumping operations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, size, and class of pumping required (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2).

d. Whether the impeller shaft sleeve will be of aluminum-bronze or corrosion-resistant steel (see 3.10.4).

e. Whether a gasoline or diesel engine is required (see 3.11).

f. Whether the pumping units shall be skid mounted or wheel mounted (see 3.12).



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g. Color required (see 3.19).

h. When Government-loaned property is required (see 3.21).

i. When rail humping test is required (see 3.22).

j. When a check valve is not required (see 3.23.2.1).

k. Degree of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required the item shall be examined and should be a first produced pumping unit. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and approval of the documents first article.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.21, when applicable (see 6.2).

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pumping unit.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of pumping units.

6.7 Noise limits. When the noise level of the pumping unit exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the pumping unit, the distance from the pumping unit at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

6.8 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the pumping units covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.9 Definitions.

6.9.1 Cross-country operation. Cross-country operation consists of traveling over open fields, broken ground, loose sand or gravel, and uneven terrain in general.



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6.9.2 Unimproved road operation. Unimproved road operation consists of traveling over level or rolling gravel roads.

6.9.3 Highway operation. Highway operation consists of traveling over level or rolling paved roads.

6.10 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Air Force - 99

Review activity:
Navy - MC
Air Force - 84

Preparing activity:
Army - ME

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X-3045A

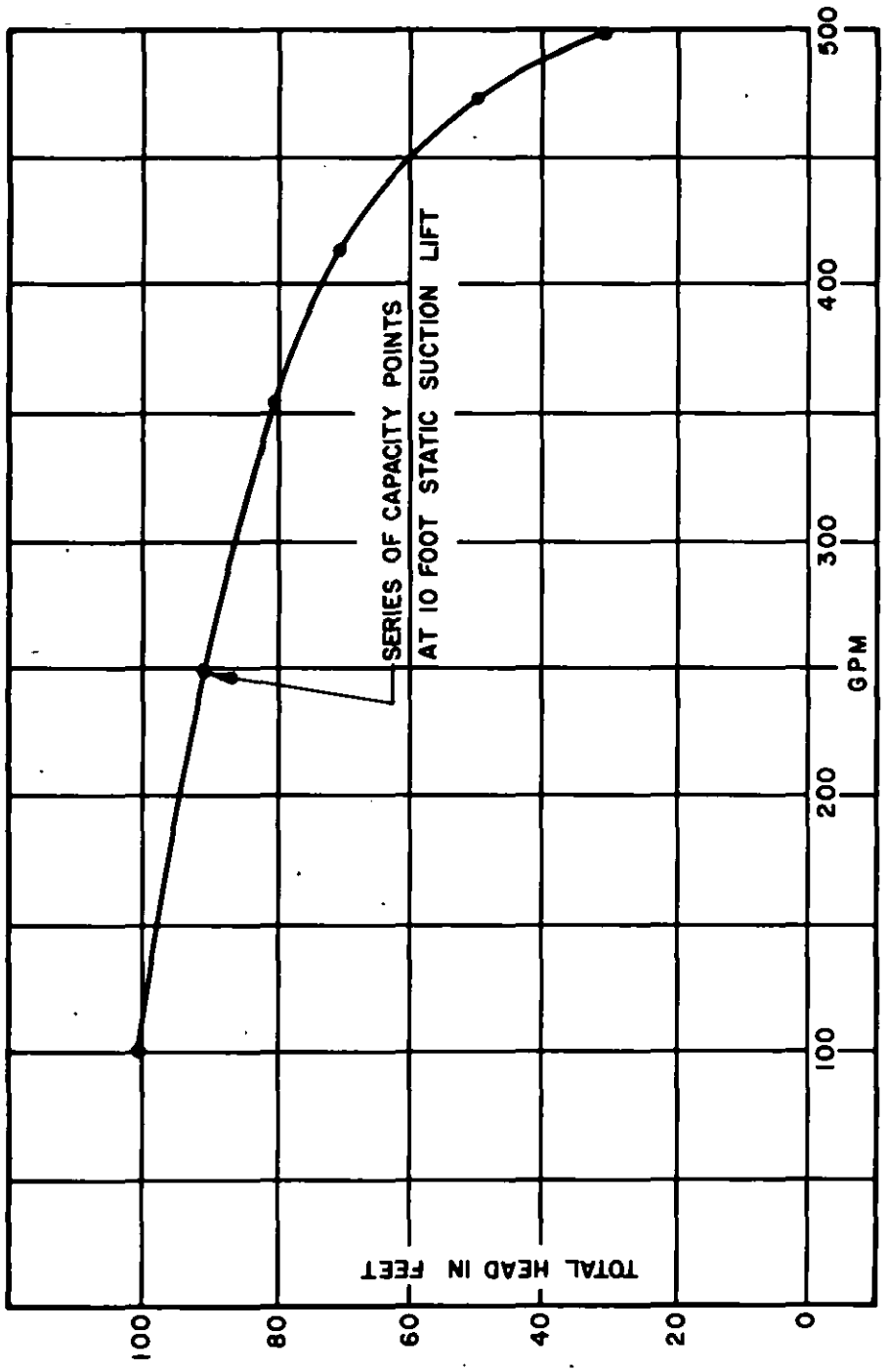



FIGURE 1. Pump, type I, size 4, class A.



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X-3046A

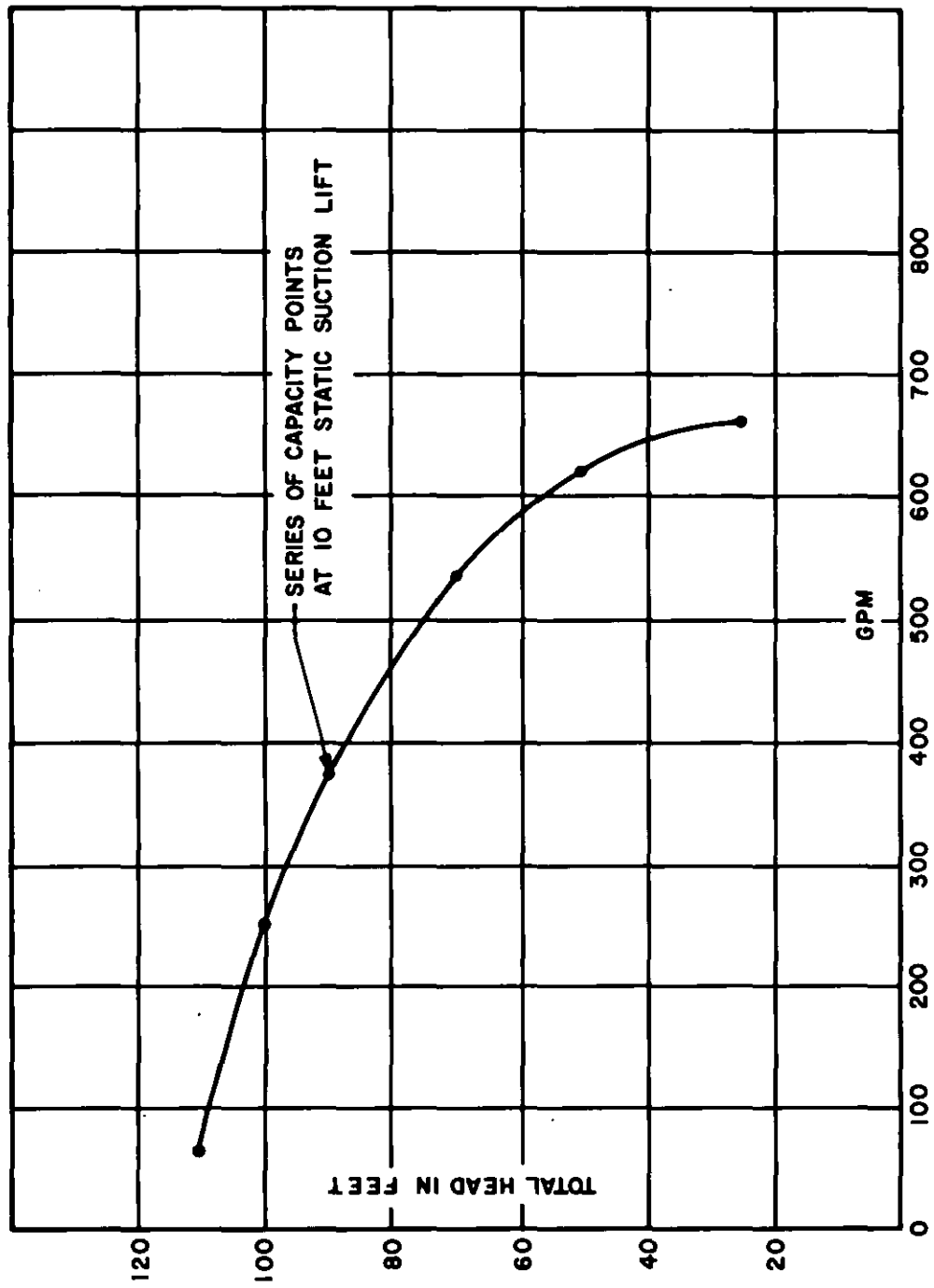
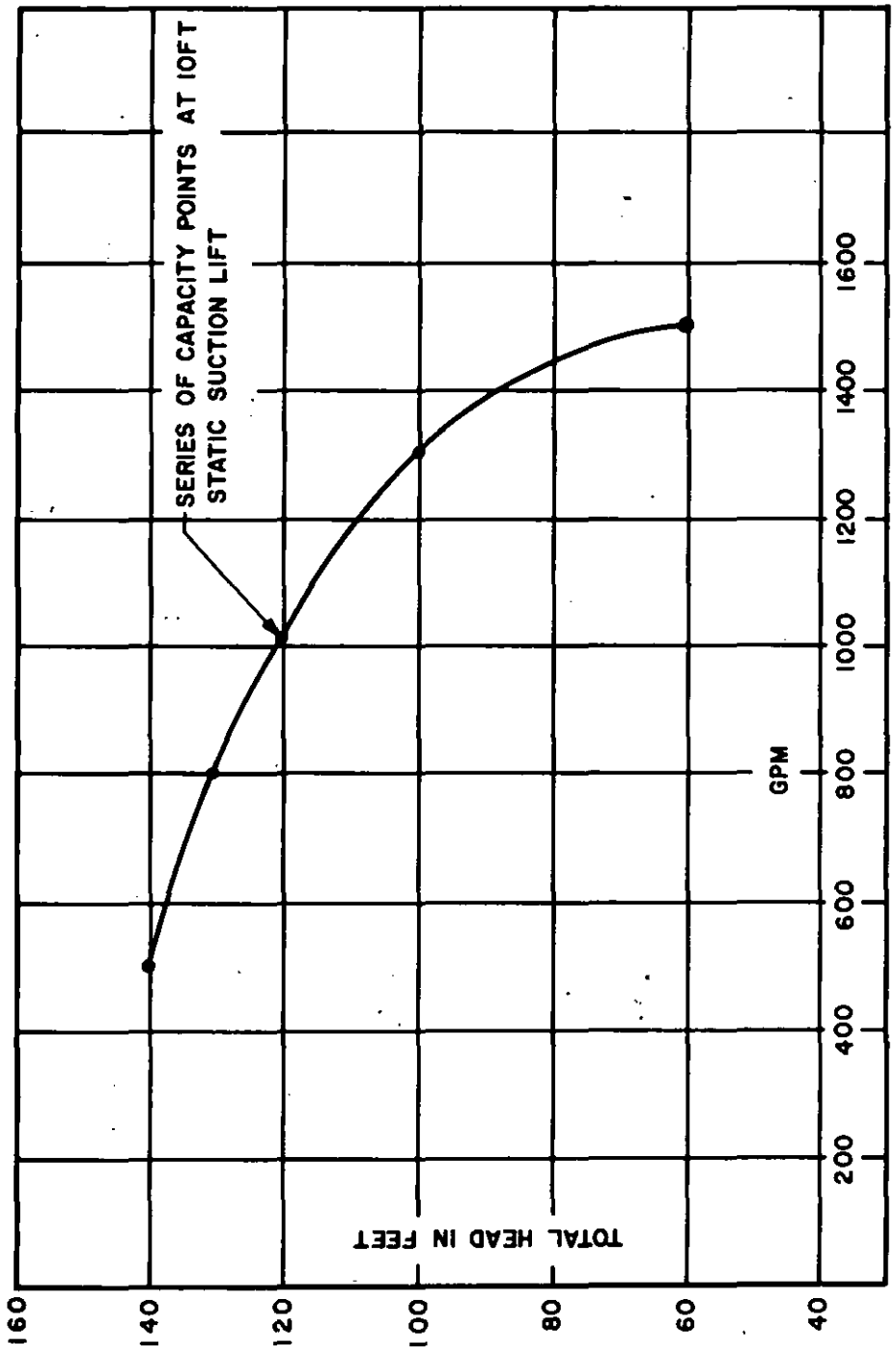



FIGURE 2. Pump, type I, size 4, class B.



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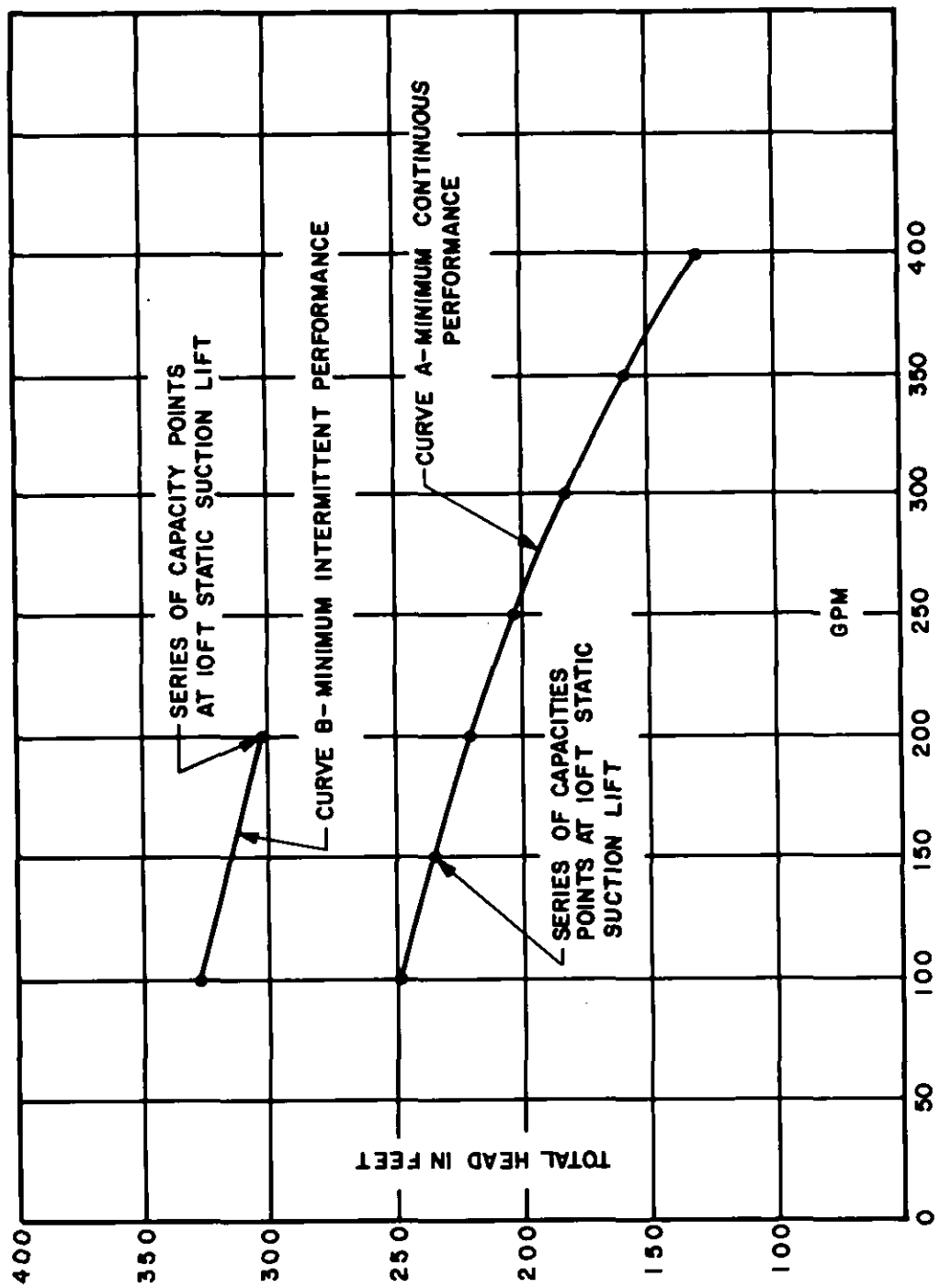



X-3047A

FIGURE 3. Pump, type I, size 6.



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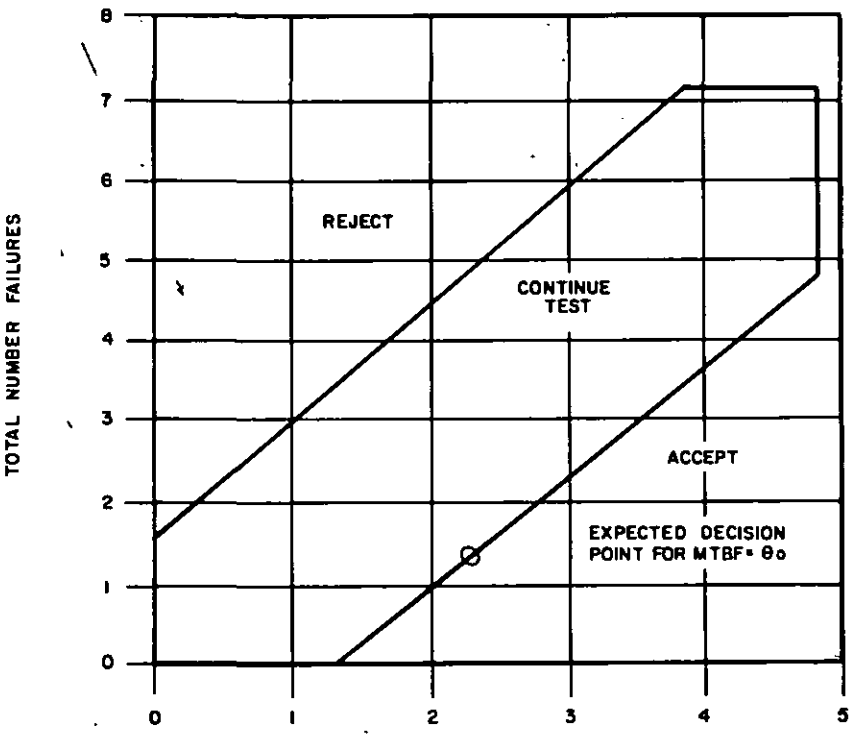

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FIGURE 4. Pump, type II, size 4.



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DECISION RISKS 20%
DISCRIMINATION RATIO 2.0:1




| NO. OF FAILURES | TOTAL TEST TIME * REJECT (EQUAL OR LESS) | ACCEPT EQUAL OR MORE |
|---|---|---|
| 0 | N/A | 1.40 |
| 1 | N/A | 2.09 |
| 2 | .35 | 2.79 |
| 3 | 1.04 | 3.48 |
| 4 | 1.73 | 4.17 |
| 5 | 2.43 | 4.87 |
| 6 | 3.12 | 4.87 |
| 7 | 3.81 | 4.87 |
| 8 | 4.87 | N/A |

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 5. Accept-reject criteria.

X-3661B



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-P-10360F | Pumps, Centrifugal: 4-Inch and 6-Inch, Gasoline-Engine- and Diesel-Engine-Driven, for Water |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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